Serag Eldien Mahmoud Abdelkarim

Backend Software Engineer | Node.js · NestJS · TypeScript · PostgreSQL · AWS

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SUMMARY

Junior Backend Software Engineer IBM Certified with 2+ years of hands-on experience building and deploying production web applications with Node.js, NestJS, and TypeScript. Experienced in designing REST APIs, authentication systems, payment integrations, database schemas, Docker-based deployments, and AWS services. Built and shipped 5+ client applications end to end, with additional experience in Spring Boot and AI-powered backend systems.

SKILLS

Programming Languages: TypeScript, JavaScript, SQL, Java, C++

Backend: Node.js, NestJS, REST API design, Spring Boot, Jest

Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM, TypeORM

Cloud & DevOps: Docker, AWS (EC2, S3), GitHub Actions (CI/CD), Nginx, Linux

Messaging & Eventing: RabbitMQ, BullMQ

Developer Tooling: Cursor, Claude Code (AI-assisted development), Git / Github, Swagger

Frontend: React, Tailwind, Tanstack router

Engineering: Clean Architecture, SOLID, Design Patterns, OOP, Unit Testing, Agile, DSA (220+ LeetCode)

Personality: Self-motivated, Adaptable, Collaborative

EXPERIENCE

Freelance Software Developer — Self-Employed (Remote) Jan 2024 – Present

Designed, built, and shipped 4+ full-stack web applications for clients end to end.

- Delivered production backends for 5+ client projects using NestJS, React, PostgreSQL, MongoDB, and AWS — cutting delivery by 3x faster delivery and shipping time by leveraging Cursor and Claude Code.
- Eliminated client-reported auth issues by implementing JWT, OAuth (Google/Facebook via Passport), OTP, Clerk , and RBAC across all projects.
- Designed relational and document schemas optimized for read-heavy query patterns; applied indexing strategies that visibly improved query performance on the Englishom platform.

AI & ML Intern — IMPACT at Zewail University Jan 2023 – Sep 2023

- Built an AI-driven document processing system in Python with OpenAI APIs, automating data extraction across PDFs — directly became the foundation for the DocTalker graduation project.
- Gained hands-on experience with cloud deployment and database-backed AI pipelines, later applied in production.

Instructor — Microsoft Student Partner (MSP) 2021 – 2022

- Taught Problem-Solving Level 0 and Level 1 using C++, covering core algorithms and data structures fundamentals for university students.
- Prepared 40+ students for exams through structured practice sets and mock contests on vJudge, tracking progress across weekly problem sets.

PROJECTS

Englishom — Online Courses Platform

englishom.com · NestJS, TypeScript, MongoDB, Docker, AWS S3, SES, Paymob, Passport OAuth

github.com/itzSerag/Englishom · Englishom Backend on Github – differ from the production repo

- Built the full NestJS + TypeScript backend for a live e-learning platform using modular monolithic architecture with barrel pattern for maintainability.
- Integrated OTP (Brevo), Google/Facebook OAuth (Passport), and Paymob payment gateway — covering the complete user auth and purchase lifecycle.
- Served course content via AWS S3 + CDN for scalable, fast media access.

DocTalker — AI RAG Document Chat Platform

github.com/itzSerag/DocTalker · Node.js, Express.js, MongoDB, OpenAI API, AWS S3, Google OAuth

- Built a RAG backend handling PDF and handwritten document uploads, text extraction, and chunk embedding via Hugging Face — enabling context-aware Q&A over user documents using OpenAI and Google Gemini
- Implemented tiered subscription system with Stripe (checkout sessions, webhook-driven access control, per-tier limits) and extended ingestion to web scraping and YouTube transcript extraction.

Chatty — Real-Time Chat Application

github.com/itzSerag/realtime_MERN_chatty-backend · NestJS, MongoDB, Socket.IO, Cloudinary, React, Tailwind CSS

- Implemented repository pattern in NestJS with MongoDB/Mongoose; built real-time messaging with Socket.io and Cloudinary media storage.
- React + Tailwind frontend with Zod-validated forms, centralized logging and exception handling for production monitoring.

EDUCATION

B.Sc. in Computer Science — Ain Shams University, Cairo, Egypt Sep 2020 – Sep 2024

- Relevant coursework: Databases, Algorithms & Data Structures, OOP, Parallel Programming, AI Intro, Java, C++
- Competed in **ECPC** (Egyptian Collegiate Programming Contest) 2022, applying algorithms and problem-solving under competition conditions.

CERTIFICATIONS

- IBM Web Development Professional Training — IBM / Credly
- IBM Software Developer Training — IBM / Credly
- Certifications & References — Certifications & Achievements

ADDITIONAL & LANGUAGES

Languages: Arabic (Native) · English (Professional Working Proficiency) **| Military Service:** Completed -- Sep 2026